

ActuaryGPT — Underwriting Report

Confidential actuarial risk assessment & premium recommendation report.

1. Customer & Policy Profile

Age (Normalized)	35.0	Gender	Male
Height (cm)	175.0	Weight (kg)	70.0
BMI	22.86	Product Info 2	D3
Occupation	Engineer	Annual Income	\$95,000.00
Insurance Type	Life	Coverage Amount	\$500,000.00

2. Risk Analysis & Premium Pricing

Predicted Risk Class	Class 1 of 8
Risk Category	Low Risk
Recommended Annual Premium	\$10,000.00

3. AI Actuarial Analysis

Actuarial Report: Insurance Application Evaluation

Applicant Details:

* **Age:** 35.0
* **BMI:** 22.86 (Healthy)
* **Smoker:** No
* **Occupation:** Engineer
* **Income:** \$95,000.00
* **Coverage Amount:** \$500,000.00 (Life Insurance)
* **Previous Claims:** 0
* **Family History:** 0 (No adverse history)
* **Lifestyle:** Exercises (1), No Alcohol (0)
* **ML Predicted Risk Class:** 1 (Lowest Risk)
* **ML Prediction Confidence:** 65.3%
* **Calculated Annual Premium:** \$10,000.00

1. Risk Assessment

The applicant presents a highly favorable risk profile across multiple key indicators:

* **Age:** At 35, the applicant is in a low-risk age bracket for life insurance.
* **BMI:** A BMI of 22.86 is well within the healthy range, indicating good physical health.
* **Smoking Status:** The applicant is a non-smoker, significantly reducing mortality and morbidity risks.
* **Occupation:** "Engineer" is typically classified as a low-risk occupation with minimal exposure to occupational hazards.
* **Previous Claims & Family History:** No previous claims and no adverse family history further contribute to a clean risk profile.
* **Lifestyle:** Positive indicators with regular exercise and no alcohol consumption point towards a healthy lifestyle, which generally translates to lower health risks.
* **Coverage Amount vs. Income:** A coverage amount of \$500,000 for an income of \$95,000 (approx. 5.26x income) is within reasonable and typical underwriting guidelines for life insurance, indicating financial justification for the coverage.

Overall, the applicant demonstrates a very low-risk profile, aligning with the Machine Learning classifier's prediction of Risk Class 1.

2. Comparative Analysis

The current application's predicted Risk Class 1 and premium of \$10,000.00 are compared against the provided similar approved historical cases:

* **Application POL-2626 (customer_9):**

* Similarity: 42.6%
* Risk Class: 2
* Premium: \$12,000.00
* **Comparison:** The current applicant exhibits a superior risk profile (Risk Class 1 vs 2) compared to POL-2626. Consequently, the proposed premium of \$10,000.00 is appropriately lower than the \$12,000.00 premium of POL-2626, reflecting this reduced risk.

* **Application POL-25755 (customer_4):**
* Similarity: 39.9%
* Risk Class: 5
* Premium: \$20,000.00
* **Comparison:** This historical case had a significantly higher risk class (5). The current applicant's Risk Class 1 is substantially better, justifying a much lower premium of \$10,000.00 compared to \$20,000.00.

* **Application POL-31852 (customer_5):**
* Similarity: 38.6%
* Risk Class: 7
* Premium: \$30,000.00
* **Comparison:** This case represents a very high-risk profile (Risk Class 7). The current applicant's extremely low risk profile (Risk Class 1) stands in stark contrast, making the \$10,000.00 premium highly appropriate relative to the \$30,000.00 charged for a much riskier individual.

In summary, the current applicant possesses a demonstrably lower risk profile than all provided similar approved historical cases. The proposed premium and risk class are consistent with, and even superior to, these approved benchmarks.

3. Anomaly & Fraud Detection

A thorough review of the applicant's details reveals no significant anomalies or indicators of potential fraud:

* **BMI:** Healthy range, not unusual.
* **Coverage vs. Income:** The \$500,000 coverage is reasonable for an income of \$95,000, not appearing excessive or speculative.
* **Claims/History:** Absence of previous claims or adverse family history is positive, not suspicious.
* **Lifestyle Indicators:** The exercise and non-alcohol consumption details align with a healthy profile, consistent with other low-risk indicators.
* **ML Prediction Confidence:** While 65.3% confidence is not extremely high, it is a solid majority and does not suggest uncertainty to the point of being an anomaly, especially given the uniformly positive risk factors.

There are no red flags that warrant further investigation for fraud or significant data inconsistency.

4. Premium Recommendation

The calculated annual premium of \$10,000.00 is well-justified:

* **Risk Class 1:** The Machine Learning model has assigned the lowest possible risk class (1), indicating a minimal risk exposure for the insurer. A premium at the lower end of the pricing spectrum is expected for such a profile.

* **Prediction Confidence:** A confidence level of 65.3% provides reasonable assurance in the model's low-risk assessment.

* **Historical Comparison:** The proposed premium of \$10,000.00 is lower than all three comparable approved historical cases, which had higher risk classes (2, 5, 7) and correspondingly higher premiums (\$12,000.00, \$20,000.00, \$30,000.00). This demonstrates that the current premium is competitive and accurately reflects the applicant's superior risk profile.

Based on the exceptionally strong risk factors and favorable comparison to approved policies, the \$10,000.00 premium is appropriate and actuarially sound for this low-risk applicant.

5. Underwriting Decision Recommendation

Recommendation: Approve

Justification:

The applicant presents an overwhelmingly strong and low-risk profile, supported by:

- * Excellent demographic and health indicators (age, healthy BMI, non-smoker).
- * Positive lifestyle choices (exercise, no alcohol).
- * Absence of adverse history (no claims, no adverse family history).
- * A low-risk occupation and appropriate coverage amount relative to income.
- * The Machine Learning model's prediction of Risk Class 1 (lowest risk).
- * A proposed premium that is consistent with, and superior to, similar approved historical cases.
- * No identified anomalies or fraud indicators.

Given the comprehensive assessment, this application represents a favorable risk for the company.

4. Sign-off & Recommendation

Based on the predictive model output and AI underwriting guidelines, the premium recommendation is formally calculated at **\$10,000.00** per annum. The underwriting team should perform secondary manual verification if the risk class exceeds Class 5.